Technical Document #667: Data Engineering - Comprehensive Overview

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Data pipelines automate the movement and transformation of data between systems. Apache Airflow and Prefect are popular workflow orchestration tools.

Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Key Takeaways:

- Schema evolution handles changes in data structure over time.
- Stream processing handles continuous data streams in real-time.
- Data lakes store raw data in its native format.